

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

EXTRACTIONTEK SALES LLC,
Petitioner,

v.

GENE POOL TECHNOLOGIES, INC.,
Patent Owner.

IPR2022-00832
Patent 9,145,532 B2

Before JEFFREY N. FREDMAN, CYNTHIA M. HARDMAN, and
JAMIE T. WISZ, *Administrative Patent Judges*.

HARDMAN, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

This is a Final Written Decision in an *inter partes* review challenging the patentability of claims 1–19 of U.S. Patent No. 9,145,532 B2 (“the ’532 patent,” Ex. 1001). We have jurisdiction under 35 U.S.C. § 6.

Petitioner has the burden of proving unpatentability of the challenged claims by a preponderance of the evidence. 35 U.S.C. § 316(e). Having reviewed the parties’ arguments and cited evidence, for the reasons discussed below, we find that Petitioner has not demonstrated by a preponderance of the evidence that claims 1–19 are unpatentable.

A. *Procedural History*

Petitioner ExtractionTek Sales LLC filed a Petition requesting *inter partes* review of claims 1–19 of the ’532 patent. Paper 1 (“Pet.”). Patent Owner Gene Pool Technologies, Inc. filed a Preliminary Response. Paper 11. In view of the then-available preliminary record, we instituted an *inter partes* review. Paper 15 (“Inst. Dec.”).

After institution, Patent Owner filed a Response to the Petition. Paper 22 (“PO Resp.”). Petitioner filed a Reply. Paper 23 (“Reply”). Patent Owner filed a Sur-reply. Paper 26 (“Sur-reply”).

On July 31, 2023, we held an oral hearing, the transcript of which is of record. Paper 31 (“Tr.”).

B. Real Parties in Interest

The parties each identify themselves as the real parties in interest.
Pet. 2;¹ Paper 7, 1.

C. Related Matters

Petitioner states that the '532 patent is among several patents asserted in *Gene Pool Technologies, Inc. v. ANM, Inc.*, Case No. 1:21-cv-01154-CL (D. Oregon). Pet. 2. According to Petitioner, this case was transferred to the United States District Court for the Central District of California, where it was captioned *Gene Pool Technologies, Inc. v. ANM, Inc.*, Case No. 2:21-cv-08756-JWH-SHK, and then consolidated into *Gene Pool Technologies, Inc. v. Coastal Harvest, LLC*, Case No. 5:21-cv-01328-JWH-SHK (C.D. California). *Id.* at 2–3. Patent Owner identifies the same two California cases (2:21-cv-08756-JWH-SHK; 5:21-cv-01328-JWH-SHK). Paper 7 (Patent Owner's Mandatory Notices), 1. Petitioner is not a party in the co-pending litigation.

The parties also identify IPR2022-00625 (U.S. Patent No. 9,587,203) and IPR2022-01011 (U.S. Patent No. 9,144,751). Pet. 3; Paper 13 (Petitioner's Updated Mandatory Notices), 1; Paper 7 (Patent Owner's Mandatory Notices), 1.

Patent Owner identifies related U.S. Patent Nos. 9,682,333, 9,926,513, 10,595,555, and 10,974,165. Paper 7 (Patent Owner's Mandatory Notices), 2.

¹ Petitioner states that it has “two trade names: ExtractionTek Solutions and ExtractionTek Stainless.” Pet. 2.

D. The '532 Patent (Ex. 1001)

The '532 patent relates to systems and methods for extracting solute from source materials. Ex. 1001, code (54), 1:6–7. According to the Specification, “many existing systems are configured to extract from a single container of source material,” and are thus “unable to simultaneously extract from materials stored in a plurality of distinct containers,” leading to a “bottleneck.” *Id.* at 1:11–16. “[M]any conventional extraction systems do not allow extraction to be performed in a single, closed loop process that reclaims solvent and re-introduces the reclaimed solvent in subsequent cycles of the system,” and manual reclaiming and reuse of solvent is inefficient. *Id.* at 1:22–28. The Specification identifies “a need for cooling reclaimed solvent to a liquid state prior to collect[ing]” the solvent, because “[l]iquid solvent may be more efficiently stored” than gaseous solvent. *Id.* at 1:41–44.

Figure 1 of the '532 patent is reproduced below.

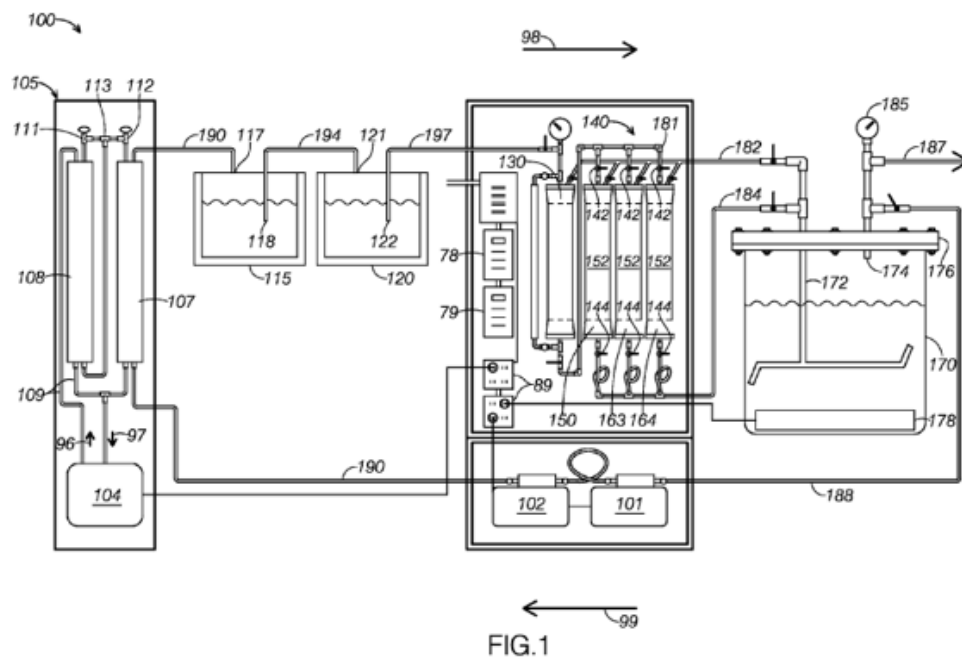


Figure 1, reproduced above, depicts a system for extracting solute from a source material. *Id.* at 2:34–35. The system includes solvent source container 120, solvent compressor 130, detachable canister system 140, extract container 170, first pump 101, second pump 102, condensing system 105, and solvent collection container 115. *Id.* at 3:10–14.

The Specification explains that “[s]ource container output 122 is configured to direct solvent contained in solvent source container 120 to solvent compressor 130 via a solvent source line 197.”² *Id.* at 3:66–4:1. The Specification states that “solvent source line 197 may include an internal filter,” which “may be used to remove impurities in solvent prior to introducing the solvent to detachable canister system 140.” *Id.* at 4:4–7. “Solvent compressor 130 may be used to compress solvent to an extraction pressure, the extraction pressure selected to maintain solvent in a liquid state even when exposed to an elevated extraction temperature.” *Id.* at 4:53–56.

According to the Specification, “detachable canister system 140 allows a user to refill a selected canister as one or more of the other canisters remain closed and to continue extracting solute from contained source material.” *Id.* at 5:23–26. Additionally, “solvent compressor 130 is configured to communicate with each detachable canister individually,” so that “when one canister is attached and extracting, solvent compressor 130 may charge solvent for a second canister.” *Id.* at 5:21–22, 6:20–22.

The Specification describes an example in which butane is used to extract essential oils from lavender plants contained within a canister to

² Throughout this Decision, we omit bolding of reference numbers in quotes from the challenged patent and prior art references.

produce an extract solution. *Id.* at 3:18–21. The extract mixture is heated “to a distilling temperature to produce an evaporated portion of the solvent in extract container 170,” where “[t]he distilling temperature . . . is greater than the boiling point of butane and less than typical essential oil boiling points.” *Id.* at 8:28–33. Condensing system 105 is configured to cool post-extraction solvent before the solvent reaches solvent collection container 115. *Id.* at 9:23–28.

E. The Challenged Claims

Petitioner challenges all nineteen claims of the ’532 patent. Claims 1, 9, and 14 are independent. Independent claim 1, reproduced below with bracketed letter and number designations added, is illustrative:

1. [1(a)] A method for extracting solute from a source material, the method comprising:
 - [1(b)] depositing the source material having a solute in a canister;
 - [1(c)] introducing a solvent into the canister;
 - [1(d)] exposing the source material to the solvent for a predetermined period of time to create an extract mixture having the solute in solution with the solvent;
 - [1(e)] fluidly communicating the extract mixture to a first extract container, the first extract container in fluid communication with the canister;
 - [1(f)] selectively fluidly communicating the extract mixture to a second extract container, the second extract container selectively being in fluid communication with the canister;
 - [1(g)] separating the solute from the extract mixture to define a recycled solvent by heating the one or more of the first

extract container and the second extract container to evaporate the recycled solvent; and

[1(h)] collecting the recycled solvent in a solvent collection container in fluid communication with one or more the first extract container and the second extract container.

Ex. 1001, 17:31–51.³ Independent claims 9 and 14 are similar to claim 1, but contain some different features. For example, claim 9 recites a detachable canister interface. *Id.* at 18:34–57. Claim 14 recites one extract container, whereas claim 1 recites two extract containers. *Id.* at 19:29–48.

The challenged dependent claims add additional features. For example, claims 2–8 depend directly or indirectly from claim 1 and relate to heating the extract containers and to the recycled solvent. *Id.* at 17:52–18:33. Claims 17 and 18 depend directly from claim 1 and relate to selectively removing the second extract container and placing additional source material in a second extract container. *Id.* at 20:23–44.

Claims 10–13 depend directly or indirectly from claim 9 and relate to introduction of a second canister or to connecting the canister to the detachable canister interface. *Id.* at 18:58–19:28.

Claims 15, 16, and 19 depend directly or indirectly from claim 14 and relate to, e.g., separating liquid solvent from the extract mixture, cooling the evaporated solvent, and the boiling point of the solvent. *Id.* at 20:1–22, 20:45–47.

³ For ease of reference, we adopt the letter and number designations Petitioner uses in the Petition. *See, e.g.*, Pet. 28 n.3.

F. The Asserted Grounds of Unpatentability

We instituted trial based on the following grounds of unpatentability:

Ground	Claim(s) Challenged	35 U.S.C. § ⁴	Reference(s)/Basis
1	14, 15	103	Britt, ⁵ Hebert ⁶
2	19	103	Britt, Hebert, Buese ⁷
3	16	103	Britt, Hebert, Liebert ⁸
4	1–8, 17, 18	103	Buese, Hebert, Britt, Main ⁹
5	9–12	103	Buese, Hebert, Main
6	13	103	Buese, Hebert, Main, Jones ¹⁰

Inst. Dec. 7–8, 34; Pet. 6.

Petitioner supports its contentions with a declaration from Mr. Fritz Chess (Ex. 1002), among other evidence. Patent Owner supports its

⁴ Petitioner asserts that the effective filing date of the challenged claims is November 4, 2013. Pet. 11. Although Patent Owner did not address the effective filing date in its Patent Owner Response, its declarant, Dr. Miller, was asked to assume that “the patent’s earliest asserted effective filing date [is] November 4, 2013.” Ex. 2008 (Miller Decl.) ¶ 43. For purposes of this Decision, we use November 4, 2013, as the effective filing date. *See* Ex. 1001, code (22).

⁵ Britt, U.S. Patent Pub. 2006/0041154 A1, published February 23, 2006 (“Britt,” Ex. 1004).

⁶ Hebert et al., U.S. Patent No. 5,516,923, issued May 14, 1996 (“Hebert,” Ex. 1009).

⁷ Buese et al., U.S. Patent No. 9,242,189 B2, issued January 26, 2016 (“Buese,” Ex. 1005).

⁸ Liebert, U.S. Patent No. 3,267,689, issued August 23, 1966 (“Liebert,” Ex. 1007).

⁹ Main, U.S. Patent No. 2,457,251, issued December 28, 1948 (“Main,” Ex. 1006).

¹⁰ Jones, U.S. Patent No. 5,074,332, issued December 24, 1991 (“Jones,” Ex. 1008).

contentions with a declaration from Gregory P. Miller, Ph.D. (Ex. 2008), among other evidence.

Based on their statements of qualifications and curricula vitae, and in view of the discussion below in Section II.A.2 regarding Dr. Miller's qualifications, we find that Petitioner's declarant Mr. Chess, and Patent Owner's declarant Dr. Miller, are qualified to provide technical opinions from the perspective of a person of ordinary skill in the art in this proceeding. Ex. 1002 (Chess Decl.) ¶¶ 2–5, pp. 69–70 (Mr. Chess's statement of qualifications and curriculum vitae); Ex. 2008 (Miller Decl.) ¶¶ 4–12, pp. 1–11 (appended after Declaration signature page) (Dr. Miller's statement of qualifications and curriculum vitae).

II. ANALYSIS

A. *Level of Ordinary Skill in the Art*

1. *Defining the Level of Ordinary Skill in the Art*

We consider the grounds of unpatentability in view of the understanding of a person of ordinary skill in the art as of the effective filing date of the challenged claims. *See, e.g., Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). Petitioner contends that a person of ordinary skill in the art (sometimes referred to herein as a “POSITA” or “POSA”) would typically have either:

- (1) a bachelor's degree in mechanical engineering, chemical engineering, petroleum engineering, or an equivalent field; and some academic or industry experience working with, designing, or studying methods of extracting solute from a source material or similar devices; or
- (2) at least four years of industry experience working with, designing, or studying methods of extracting solute from a source material or similar devices.

Pet. 11 (citing Ex. 1002 (Chess Decl.) ¶ 13).

Patent Owner “propose[s] to expand the level of ordinary skill in the art” recited in Petitioner’s proposal “to include educational experience in chemistry, biology, or equivalent fields of physical or life science,” as follows (Patent Owner’s proposed additions are shown in bold):

- (1) a bachelor’s degree in **chemistry, biology**, mechanical engineering, chemical engineering, petroleum engineering, or an equivalent field **of physical science, life science, or engineering**; and some academic or industry experience working with, designing, or studying methods of extracting solute from a source material or similar devices; or
- (2) at least four years of industry experience working with, designing, or studying methods of extracting solute from a source material or similar devices.

PO Resp. 4–5; Ex. 2008 (Miller Decl.) ¶¶ 39–43. Patent Owner contends that these additions are appropriate because “[t]he ‘educational level of the inventor’ is relevant in determining the level of skill in the art,” and one of the inventors listed on the ’532 patent has a life sciences degree. PO Resp. 4, n.1 (citing *Env’t Designs, Ltd. v. Union Oil Co.*, 713 F.2d 693, 696 (Fed. Cir. 1983)); Ex. 2008 (Miller Decl.) ¶ 42.

On the full trial record, we adopt Petitioner’s proposed level of ordinary skill in the art, with Patent Owner’s additions. The additions are consistent with the evidence of record. For example, we find that adding chemistry and biology degrees is appropriate in view of Dr. Miller’s testimony that college-level chemistry and biology courses typically include training on extractive technologies. Ex. 1021 (Miller Tr.), 38:9–39:18. We also agree with Patent Owner that the educational level of the inventors may be considered (*see Env’t Designs, Ltd.*, 713 F.2d at 696), and on this record,

we accept Patent Owner and Dr. Miller’s undisputed representation that “named inventor Jason Wasserman had a bachelor’s degree in Agricultural Science and Sustainable Agriculture, a life sciences degree.” Ex. 2008 (Miller Decl.) ¶ 42; PO Resp. 4.

Thus, for the reasons above, we define a person of ordinary skill in the art as someone who would have had

- (1) a bachelor’s degree in chemistry, biology, mechanical engineering, chemical engineering, petroleum engineering, or an equivalent field of physical science, life science, or engineering; and some academic or industry experience working with, designing, or studying methods of extracting solute from a source material or similar devices; or
- (2) at least four years of industry experience working with, designing, or studying methods of extracting solute from a source material or similar devices.

2. *Whether Dr. Miller Qualifies as a Person of Ordinary Skill in the Art*

Petitioner argues that Dr. Miller does not qualify as a person of ordinary skill in the art because “he lacks the specific academic or industry experience working with, designing, or studying methods of extracting solute from a source material or similar devices.” Reply 3. Petitioner argues that “between 1988–2015 [Dr. Miller] performed no organic butane extraction, organic propane extraction, or CO₂ organic material extraction.” *Id.* (citing Ex. 1021 (Miller Tr.), 58:7–18). Petitioner further argues that between 1988 and 2015, “the sole organic closed-loop extraction” Dr. Miller performed was “Soxhlet-style extraction,” which Petitioner contends is “rudimentary and provides insufficient basis in establishing one’s standing as a PHOSITA.” *Id.* at 4 (citing Ex. 1021 (Miller Tr.), 59:4–16).

We are not persuaded that Dr. Miller does not qualify as a person of ordinary skill in the art. As an initial matter, we note that Petitioner is improperly trying to narrow its original proposed level of skill in the art to now require specific experience with “organic extractions” and/or “closed loop extraction systems.” *See* Reply 5 (“Dr. Miller cannot testify to the knowledge of one skilled in the art at the time of the ‘532 Patent because as of 2013, Dr. Miller had never performed organic extractions.”), 6 (criticizing Dr. Miller’s experience because “his experience and expertise in closed loop extraction systems began only after 2015”). Petitioner’s original proposal was much more broadly drawn, i.e., to “experience working with, designing, or studying methods of extracting solute from a source material.” Pet. 11. As explained in our Trial Practice Guide, in a reply brief, a party is not permitted to “proceed in a new direction with a new approach as compared to the positions taken in a prior filing,” as Petitioner attempts to do here. Trial Practice Guide,¹¹ 74; *see also* 37 C.F.R. § 42.23(b). We do not consider Petitioner’s belated arguments seeking to narrow its broad proposal to require experience with “organic extractions” or “closed loop extraction systems.”

Petitioner acknowledges that “Dr. Miller has advanced degrees in life sciences,” which comports with the educational requirement specified in part (1) of the level of skill we use herein. Reply 3; *supra* Section II.A.1 (referring to “a bachelor’s degree in . . . **biology**, . . . or an equivalent field of . . . **life science**”). Additionally, Dr. Miller has “decades (1988 to present) of

¹¹ PTAB Consolidated Trial Practice Guide (Nov. 2019) (“Trial Practice Guide”), *available at* <https://www.uspto.gov/TrialPracticeGuide> Consolidated.

direct experience in the use of partial and complete extractions of diverse materials in environmental studies.” Ex. 2008 (Miller Decl.) ¶ 8. He has also “studied, used, and developed thousands of extractions to solubilize and remove oils, wastes, metals, and other toxins from soil, water, plants, and animal tissues for analysis.” *Id.* He avers that “[e]xtractive chemistry was core to [his] business practice in 1988 and remains so to this day,” and that “[b]y 1992 [he] would have developed ordinary skill in the art having 4 years of experience relevant to the current subject.” *Id.* ¶ 10.

Based on this experience, we determine that Dr. Miller qualifies as a person of ordinary skill in the art because by 1992, he had “at least four years of industry experience working with, designing, or studying methods of extracting solute from a source material or similar devices,” as required by the level of ordinary skill we adopt herein. He also avers that he gained additional training and engaged in additional extraction work prior to the effective filing date. *Id.*

Additionally, between 2015 and 2020, Dr. Miller “enter[ed] the cannabis extraction and concentration industry, designing, building, and operating [his] own facility,” further qualifying him as a person of ordinary skill in the art. Ex. 2008 (Miller Decl.) ¶ 10; *see also* Reply 6 (acknowledging Dr. Miller’s post-2015 “experience and expertise in closed loop extraction systems”). Although this work started in 2015 (two years after the effective filing date of 2013), a person of ordinary skill in the art need not have been one of ordinary skill at the time of a patent’s filing date. *See U.S. Endoscopy Group, Inc. v. CDX Diagnostics, Inc.*, IPR2014-00639, Paper 27, at 18 (Sept. 14, 2015) (“A witness must provide testimony about

the level of skill in the art as of the critical date; however, the witness need not have acquired that knowledge as of the critical date.”).

For the reasons discussed above, we determine that Dr. Miller is qualified in this proceeding to testify from the perspective of a person of ordinary skill in the art.

B. Claim Construction

In an *inter partes* review, we interpret a claim “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b). Under this standard, we construe the claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.*

Petitioner asserts that “the claim terms should be construed according to their ordinary and customary meaning,” but also states that certain terms (i.e., “canister,” “extract container,” “solvent collection container,” and “boiling point”) “warrant clarification.” Pet. 12. Patent Owner does not take issue with these terms, and instead requests that we construe certain of the claimed method steps “as distinct, sequential, and non-continuous steps,” and that we construe the phrase “exposing . . . for a predetermined period of time.” *Id.* For completeness, we note that although neither party formally requested construction of the terms “detachable canister interface” and “support” (recited in claim 9) and “line” (recited in claim 15), both parties present claim construction arguments as to the meanings of these terms. *See, e.g.*, PO Resp. 32, 34–36, 37; Reply 22–24.

After consideration of the full trial record, we determine that in order to resolve the controversy between the parties, we need address only the

term “exposing . . . for a predetermined period of time.” *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1375 (Fed. Cir. 2019) (“The Board is required to construe ‘only those terms . . . that are in controversy, and only to the extent necessary to resolve the controversy.’”) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

1. “*exposing . . . for a predetermined period of time*”

In step [1(d)], independent claim 1 recites “*exposing the source material to the solvent for a predetermined period of time to create an extract mixture having the solute in solution with the solvent.*” Ex. 1001, 17:35–37 (emphasis added). Independent claims 9 and 14 recite a similar step. *Id.* at 18:43–44, 19:31–34.

Neither party requests a construction for this term that encompasses a specific amount of time or achieves a specific goal (e.g., exposing the source material to solvent for an amount of time sufficient to extract the desired amount of oil). Instead, Patent Owner seeks a construction of this term that excludes continuous processes. PO Resp. 6.

More specifically, Patent Owner contends that “exposing . . . for a predetermined period of time” refers to “the time a source material is exposed to a volume of solvent,” and that the Specification makes clear the exposing step is always performed in a canister that is “closed to fluid communication.” *See* PO Resp. 6–9 (citing, e.g., Ex. 2008 (Miller Decl.) ¶¶ 31–35; Ex. 1001, 3:28–34, 5:23–26, 5:58–60, 5:49–51, 13:30–33, 18:58–19:10). Patent Owner suggests that this understanding of the term limits the claims to batch processes (and excludes continuous processes). *See, e.g.*, PO Resp. 8 (“The ’532 patent continues to describe the method as a batch

process and never suggests that its method involves a continuous extraction process.”).

Petitioner disputes Patent Owner’s proposed construction, and advocates that “exposing” and “predetermined period of time” should “be given their plain and ordinary meaning.” Reply 7–8. Petitioner notes that in the co-pending litigation, the parties agreed that “predetermined period of time” means “period of time determined in advance,” and the court issued a claim construction order that includes this agreed-upon construction. *Id.* at 7 (citing Ex. 3001, 11).

We agree with Petitioner that “exposing . . . for a predetermined period of time” should “be given [its] plain and ordinary meaning.” Reply 7–8. To the extent the plain and ordinary meaning needs elaboration, we adopt the agreed-upon construction of “predetermined period of time” from the related district court litigation, i.e., a “period of time determined in advance.”¹² Ex. 3001, 11.

For clarity, we note that nothing in this plain and ordinary meaning limits the term to a batch process or excludes a continuous process. We do not see any basis to read into this limitation requirements that the source material contact a volume of solvent within a closed or sealed container. *See* Sur-reply 4. Although the Specification indicates that the step of exposing the source material to the solvent for a predetermined period of time *may* take place in a canister that is “closed to fluid communication,” Patent

¹² Patent Owner appears to acquiesce to this construction in this case, stating that “Patent Owner is not proposing a different construction of the limitation ‘predetermined period of time,’ which the parties to the District Court action agreed means ‘period of time determined in advance.’” Sur-reply 3–4.

Owner has not pointed us to any persuasive intrinsic evidence indicating that the exposing step *must* take place in a canister that is “closed to fluid communication.” The Specification “uses the broad permissive term ‘may’ when describing a ‘user *may* use the output valves to direct the extract mixture contained in an associated canister to extract container.’” Reply 10 (quoting Ex. 1001, 5:37–39, emphasis Petitioner’s). Accordingly, Patent Owner does not persuade us that the intrinsic record “strongly support[s] that ‘exposing the source material to the solvent for a predetermined period of time to create an extract mixture’” requires a batch process or excludes a continuous process. *See* PO Resp. 6.

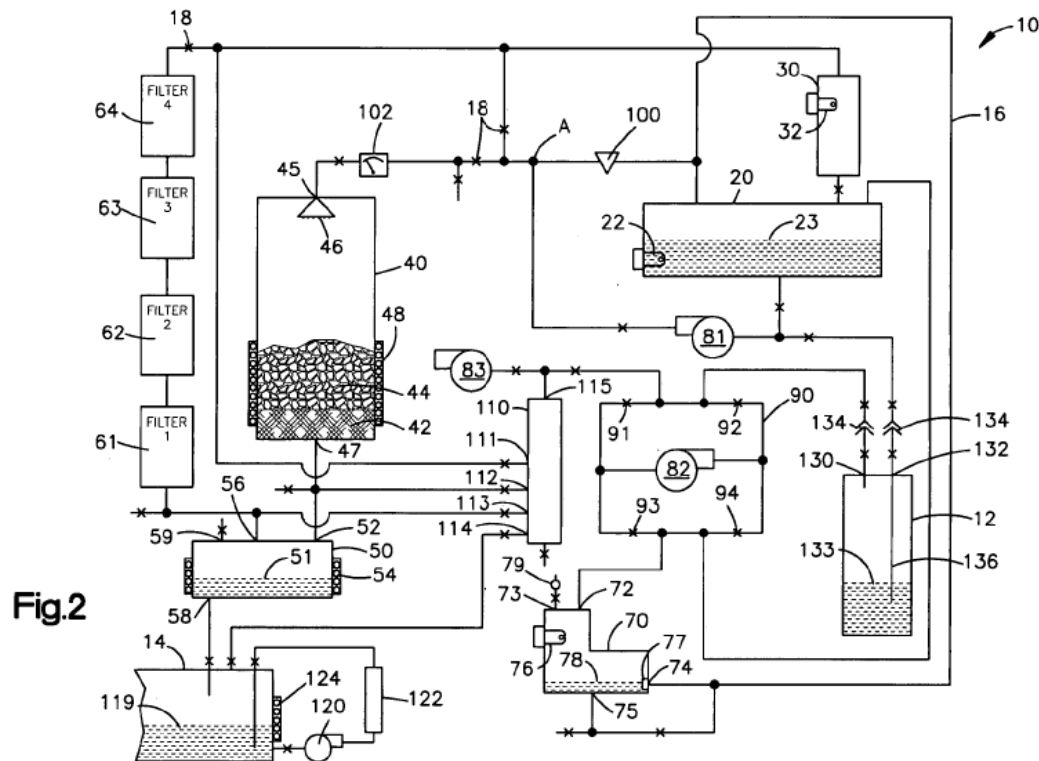
2. *Conclusion*

We determine that “exposing . . . for a predetermined period of time” should be given its plain and ordinary meaning, i.e., “exposing . . . for a period of time determined in advance.”

C. *Overview of Asserted Prior Art References*

1. *Britt (Ex. 1004)*

Britt, titled “Oil Extraction Method,” “relates to methods for extracting oils from oil bearing materials.” Ex. 1004, code (54), ¶ 2. We reproduce Britt Figure 2 below.



Britt Figure 2 is a schematic diagram of an oil extraction system, which can be used to extract oil from plant materials. *Id.* ¶¶ 5–6, 10.

During system operation, plant materials such as rose petals 44 are placed in extraction tank 40. *Id.* ¶¶ 13, 23. Solvent 23 is pumped from reservoir 20 through the rose petals to extract rose oil and yield a mixture of solvent and oil. *Id.* ¶¶ 12, 13. Distillation tank 50 receives the solvent/oil mixture from extraction tank 40. *Id.* ¶ 14. The mixture is heated “to a temperature that is high enough to evaporate the solvent, but low enough to avoid decomposing or evaporating the oil.” *Id.* ¶¶ 28, 14. Evaporated solvent vapor flows out of distillation tank 50, leaving the oil in distillation tank 50. *Id.* ¶¶ 15, 28. The solvent continues through filter tanks 61–64 to remove particles, water, and acid, and then flows through chiller 30 back to the reservoir 20. *Id.* ¶¶ 15, 28. Chiller 30 has a cooling coil 22 for

condensing solvent vapor into a liquid and maintaining the solvent in the liquid phase. *Id.* ¶ 12.

2. *Hebert (Ex. 1009)*

Hebert, titled “Extracting Oil from Oil Bearing Plant Parts,” “relates to a method for extracting oil from oil-bearing plant parts.” Ex. 1009, code (54), 1:16–18. Hebert Figure 1 is reproduced below.

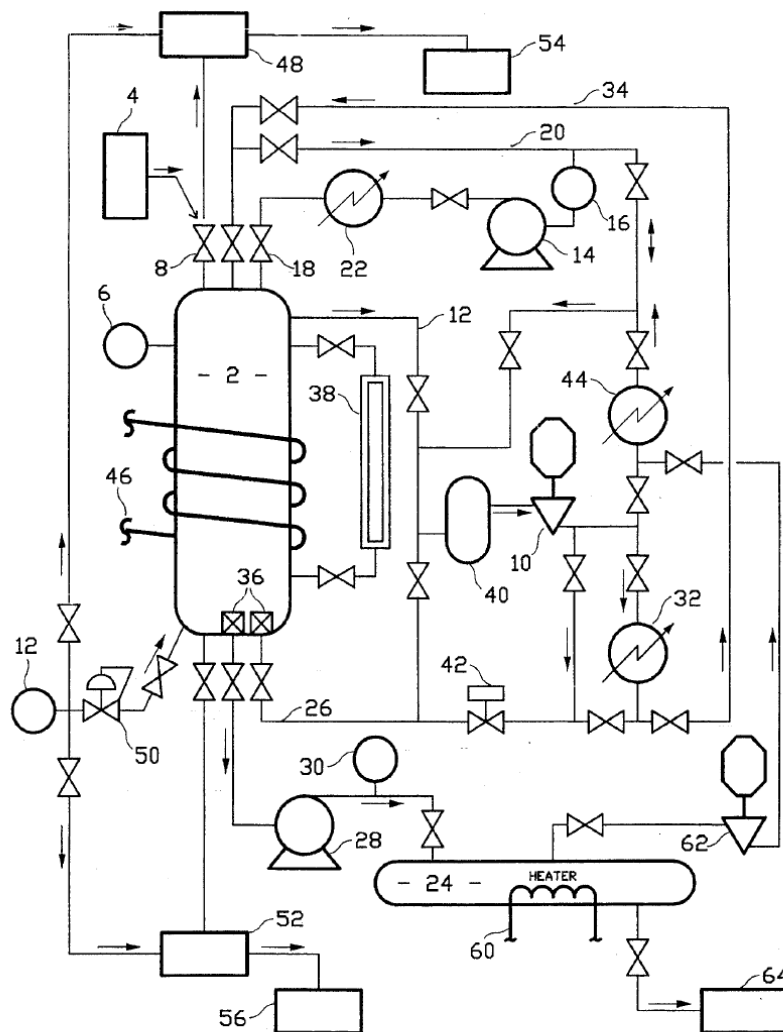


FIGURE 1

Hebert Figure 1 is a schematic illustration of a system for extracting oil from an oil-bearing plant material. *Id.* at 3:49–53. The system includes reactor

vessel 2, which is “designed to receive an oil bearing plant material, for example, rice bran.” *Id.* at 3:62–67. Liquid solvent, such as propane, is pumped by pump 14 from tank 16 into reactor 2. *Id.* at 4:25–28. The liquid solvent contacts the rice bran for a time sufficient to dissolve a substantial portion of the rice’s oil. *Id.* at 2:62–64. The liquid solvent and oil combination is transferred from reactor 2 to separator 24. *Id.* at 4:59–62. In separator 24, the solvent and oil combination is heated to vaporize the solvent, and the solvent vapor is drawn from separator 24 by compressor 62. *Id.* at 7:34–38. The solvent vapor is then liquified by condenser 44 and sent to storage tank 16. *Id.* at 7:38–39.

3. *Buese (Ex. 1005)*

Buese, titled “Continuous Extractor, Concentrator and Dryer,” relates “to a continuous immersion extraction system.” Ex. 1005, code (54), 2:18–19. Buese describes a unit that allows “the recycling of the extraction solvent, and the drying of the extracted plant material while all are fully contained within the unit.” *Id.* at 1:35–40. “The unit employs a plurality of extraction chambers such that one extraction chamber is removed and replaced without halting the extraction process in other extraction chambers.” *Id.* at 1:41–44. Buese describes the use of connectors, such as quick release connectors, for its system. *Id.* at 4:48–51.

Buese explains that “[t]he temperature at which the solvent is introduced into the extraction chambers can be controlled by employing a heater, chiller, or other heat exchanger in the solvent reservoir or the conduits situated prior to flow into the extraction chambers.” *Id.* at 5:10–14. Buese discloses butane as a solvent. *Id.* at 7:7–65.

4. *Liebert (Ex. 1007)*

Liebert, titled “High and Low Temperature Refrigeration Systems with Common Defrosting Means,” relates to a refrigeration system. Ex. 1007, 1:3–13. Liebert teaches a coolant “closed loop system” with an “expansion valve.” *Id.* at 4:9–10, 5:36–39.

5. *Main (Ex. 1006)*

Main, titled “Quick Connect Self-Sealing Coupling,” describes a quick connect coupling sleeve that advantageously enables quick coupling or uncoupling by rotating the sleeve a fractional turn relative to another part. Ex. 1006, 1:28–33, 3:15–22. The coupling is self-sealing, i.e., “the moving valve parts are effectively sealed against leakage of the operating fluid.” *Id.* at 1:5–9.

6. *Jones (Ex. 1008)*

Jones, titled “Hose Coupling Unit for Refrigerant System,” is directed to “an improved coupling unit for manually connecting a flexible hose to a process fitting on a refrigeration or air conditioning system.” Ex. 1008, 1:1–2, 1:46–50. The coupling unit is a “quick connect and disconnect coupler for locking the unit to the process fitting,” which “assures that the valve within the process fitting remains in its closed position until after the coupling unit is connected to the fitting.” *Id.* at 1:51–55. The coupling unit “assures that no refrigerant will escape through the process fitting during connecting and disconnecting the coupling unit.” *Id.* at 1:58–60.

D. *Legal Standards*

“In an [inter partes review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed.

Cir. 2016); 37 C.F.R. § 42.104(b). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review). To prevail, Petitioner must demonstrate patentability by a preponderance of the evidence. 35 U.S.C. § 316(e).

A claim is unpatentable as obvious “if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains.” 35 U.S.C. § 103; *see also KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved based on underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) any objective indicia of nonobviousness. *Graham*, 383 U.S. at 17–18. An obviousness determination requires a reason to combine “accompanied by a reasonable expectation of achieving what is claimed in the patent-at-issue.” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367 (Fed. Cir. 2016).

E. Analysis

Independent claims 1 and 9 recite a step of “exposing the source material to the solvent *for a predetermined period of time* to create an extract mixture having the solute in solution with the solvent.” Ex. 1001, 17:35–37, 18:44–46 (emphasis added). Independent claim 14 recites a similar limitation. *See id.* at 19:31–34 (“exposing the source material in a canister to a liquid solvent *for a predetermined period of time* to create an extract

mixture having the solute in solution with the liquid solvent”) (emphasis added).

Petitioner challenges independent claims 1, 9, and 14 in Grounds 1, 4, and 5, respectively. *See* Pet. 6. In each of these Grounds, Petitioner uses Hebert to teach the *predetermined period of time* claim limitation. *See id.* at 14 (for Ground 1, arguing that “Britt does not explicitly describe exposing the source material to the solvent *for a predetermined period of time* to create an extract mixture,” and that “Hebert cures this deficiency”), 25–26 (for Ground 4, arguing that Buese does not disclose the “predetermined period of time” limitation and that Hebert cures this deficiency), 41 (arguing same for Ground 5).

Patent Owner argues that Petitioner fails to explain how or why a person of ordinary skill in the art would have modified Britt or Buese with the teachings of Hebert. *See* PO Resp. 20.

After considering all of the arguments and cited evidence of record, we determine that Petitioner has not established by a preponderance of the evidence that any challenged claim is unpatentable as obvious. In particular, we agree with Patent Owner that Petitioner fails to explain how or why a person of ordinary skill in the art would have modified Britt or Buese with the teachings of Hebert. *See, e.g.*, PO Resp. 23.

In Ground 1, Petitioner maps most of the limitations of independent claim 14 to Britt. *See, e.g.*, Pet. 15–17. In Grounds 4 and 5, Petitioner maps most of the limitations of independent claims 1 and 9 to Buese. *See id.* at 28–31. Britt and Buese both teach continuous extraction systems, wherein exposing the source material to a solvent “leads immediately, inexorably, and continuously to the solvent percolating (Britt) or flowing (Buese)

through the source material and to the separation component.” Ex. 2008 (Miller Decl. ¶ 46); PO Resp. 2, 22–23. Petitioner asserts that neither Britt nor Buese “explicitly describe[s] exposing the source material to the solvent *for a predetermined period of time* to create an extract mixture.” Pet. 14, 25, 41–42. For this teaching, Petitioner turns to Hebert. *See id.*

Hebert “relates to a method for extracting oil from oil-bearing plant parts.” Ex. 1009 (Hebert), 1:16–22. In an exemplary method, Hebert discusses a system for extracting oil from rice bran. *See id.* at 3:62–67. A reactor vessel is loaded with rice bran, and liquid solvent is added. *See id.* at 4:1–2, 4:25–28. Hebert teaches that after sufficient solvent is added,

[t]he liquid solvent is constantly maintained by means of heat exchanger 22 within the temperature range of 68°–125° F., preferably 68°–72° F., prior to and during liquid filling of reactor 2 in order to cause the liquid solvent to dissolve the rice oil from the rice bran in reactor 2 ***at a predetermined rate and time.***

Id. at 4:40–45 (emphasis added). Petitioner specifically relies on the above-quoted portion of Hebert as “provid[ing] a rationale for a PHOSITA to expose the source material to the solvent for a predetermined period of time.” Pet. 14 (citing Ex. 1002 (Chess Decl.) ¶ 72; Ex. 1009 (Hebert), 4:40–45), 25, 41–42; *see also* Reply 16 (repeating same statement).

The failing in Petitioner’s arguments is that the Petition never identifies what “rationale” Hebert allegedly provides for a person of ordinary skill in the art “to expose the source material to the solvent for a predetermined period of time.” Pet. 14. We have examined each citation and argument in the Petition, but find no articulation of a reason why a person of ordinary skill in the art would have applied this teaching in Hebert

to Britt and Buese's processes. We walk through each citation and argument below.¹³

Petitioner cites Mr. Chess's declaration at paragraph 72, but that paragraph merely repeats the Petition, and thus also fails to specify a rationale for why a person of ordinary skill in the art would have imported Hebert's teachings into Britt or Buese. *See* Ex. 1002 (Chess Decl.) ¶ 72; Pet. 14.

Petitioner next contends that a person of ordinary skill in the art "would have been motivated to combine Britt with Hebert" because

both concern closed loop extraction systems and methods that use a recycled solvent to extract plant or other extractable material following similar steps of exposure to a solvent to create a solute/solvent mixture, separation of the solute and solvent, and collection of the solvent. (Exhibit 1009, Abstract; Exhibit 1005¹⁴, Abstract; Exhibit 1002, ¶73). Thus, a PHOSITA viewing Britt would have looked to Hebert (and vice versa) to inform possible improvements or alternative approaches to perform the same functions within the same general structure.

Pet. 15. The first sentence of this quotation establishes that Britt and Hebert's respective systems share some similarities, but does not proffer a reason why a person of ordinary skill in the art would have combined the specific cited teaching from Hebert with Britt. The second sentence quoted

¹³ Because Petitioner's arguments for Grounds 1, 4, and 5 follow the same structure, we analyze the Petition's arguments and citations for Ground 1, as representative of all three grounds.

¹⁴ Although Petitioner here cites Exhibit 1005 (Buese), based on the surrounding context, we assume that this is a typographical error, and that Petitioner intended to cite Exhibit 1004 (Britt).

above refers to “possible improvements or alternative approaches,” but fails to elaborate on what specific “improvements” or “alternative approaches” Hebert’s teaching offers Britt’s system.¹⁵ This quotation also cites Mr. Chess’s declaration at paragraph 73, but that paragraph in the Chess Declaration again merely repeats the Petition, and thus adds nothing to the analysis.

Finally, the Petition presents a claim chart. For the relevant limitation of independent claim 14 (designated 14(b)), the Petition provides the following:

14(b) exposing the source material in a canister to a liquid solvent for a predetermined period of time to create an extract mixture having the solute in solution with the liquid solvent;	Britt discloses a step wherein “[t]he solvent then flows through the petals 44, extracting oil from the petals 44 to yield a mixture of the solvent and the oil.” (Id., ¶13). Hebert teaches the maintenance of solvent temperature of the liquid solvent “during liquid filling of reactor 2 in order to cause the liquid solvent to dissolve the rice oil from the rice bran in reactor 2 at a predetermined rate and time.” (Exhibit 1009, 4:40–45). Accordingly, it would have been
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¹⁵ The second sentence quoted above also injects confusion in stating “vice versa.” This phrase suggests that Petitioner is also arguing that a person of ordinary skill in the art would have looked to Britt to inform possible improvements or alternative approaches to Hebert. Such an argument is even less developed in the Petition than Petitioner’s assertion that Hebert cures a deficiency in Britt.

	obvious to a PHOSITA to expose the source material of Buese ¹⁶ to a liquid solvent for a predetermined period of time to create an extract mixture having the solute in solution with the solvent. (Exhibit 1002, XIII.14(b)).
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Pet. 16–17. The table above quotes claim limitation 14(b) on the left, and presents three paragraphs of analysis on the right. The first and second paragraphs of the analysis simply quote teachings of Hebert and Britt. The third paragraph of the analysis does little more than recite the claim limitation, prefaced by the words “Accordingly, it would have been obvious to a PHOSITA to [insert quote of claim language].” This portion simply asserts a legal conclusion. This third paragraph also cites a corresponding claim chart in Mr. Chess’s declaration, but that chart is identical to the chart in the Petition, and thus adds nothing to the analysis in the Petition.

Patent Owner argues that Petitioner’s assertions as to how and why a person of ordinary skill in the art would have modified continuous extraction systems like Britt and Buese’s with a teaching from Hebert, which relates to batch processes, are cursory and unsupported. PO Resp. 2–3. We agree.

First, Petitioner never elaborates on *how* a person of ordinary skill in the art would have modified Britt or Buese to implement Hebert’s teaching of maintaining temperature of the liquid solvent in order to dissolve the oil

¹⁶ Although the Petition here recites “Buese,” based on the surrounding context, we assume that this is a typographical error, and that Petitioner intended to recite “Britt.”

from the source material “at a predetermined rate and time.” *See* Pet. 16. For example, to the extent a person of ordinary skill in the art would have made changes to Britt or Buese’s processes to implement this teaching, Petitioner fails to explain what changes a person of ordinary skill in the art would have made.

To the extent Petitioner is contending that a person of ordinary skill in the art would have “[m]odif[ied] Britt or Buese to stop the continuous flow of solvent through the source material in order to control solvent exposure for a ‘predetermined period of time,’” we agree with Patent Owner that this “would ‘fundamentally alter’ their operation,” given the different types of extraction systems in Hebert on the one hand (a batch process) and Britt and Buese on the other (continuous processes). PO Resp. 25 (quoting *Plas-Pak Indus., Inc. v. Sulzer Mixpac AG*, 600 F. App’x 755, 759 (Fed. Cir. 2015) (nonprecedential)); Ex. 2008 (Miller Decl.) ¶51. The extraction methods of Britt and Buese continuously and repeatedly wash source material with fresh solvent, whereas Hebert’s batch process extracts solute from a source material in a single exposure. *See* PO Resp. 23; Ex. 2008 (Miller Decl.) ¶ 47. Petitioner does not address the differences in the processes and how a person of ordinary skill in the art would have addressed them.

Second, Petitioner nowhere explains *why* a person of ordinary skill in the art would have seen an advantage in implementing Hebert’s teaching in the context of Britt and Buese’s processes. For example, as Patent Owner correctly notes, “Petitioner provides no particular explanation of why a POSA would apply Hebert’s teachings to Britt or Buese without undermining the benefits of these continuous extraction systems.” PO Resp. 19–20. To the extent a person of ordinary skill in the art would have seen an

advantage in applying Hebert's teachings to Britt and Buese, Petitioner does not identify it. This is not sufficient to sustain an obviousness challenge.

Petitioner's arguments find no additional support in the testimony of Mr. Chess. First, as noted above, his declaration simply parrots the arguments in the Petition and thus does not add any analysis not already discussed above. *See, e.g., TQ Delta, LLC v. CISCO Sys., Inc.*, 942 F.3d 1352, 1359 (Fed. Cir. 2019) ("conclusory expert testimony is inadequate to support an obviousness determination"); *InTouch Techs., Inc. v. VGO Commc'ns, Inc.*, 751 F.3d 1327, 1353–54 (Fed. Cir. 2014) (reversing district court's judgment of invalidity because the expert testimony "failed to provide any meaningful explanation for why one of ordinary skill in the art would be motivated to combine these references at the time of this invention"); *Xerox Corp. v. Bytemark, Inc.*, IPR2022-00624, Paper 9, 15 (PTAB Aug. 24, 2022) (precedential).

Second, we agree with Patent Owner that Mr. Chess's opinion on motivation to combine is entitled to little, if any, weight in view of his cross-examination testimony, which expressly undermined any reason to combine Britt and Hebert. *See* PO Resp. 2–3, 12. When specifically asked to explain paragraph 73 of his Declaration regarding his opinion articulating a motivation to combine Britt and Hebert, Mr. Chess responded: "I'm personally not motivated to combine anything from any of these guys together. I think they're all knuckleheads and didn't know what they were doing, to be frank." *Id.* at 24 (quoting Ex. 2009 (Chess Depo. Tr.), 185:2–6, *see also* 181:11–12 and 183:5 (directing Mr. Chess to paragraph 73 of his Declaration)). Petitioner attempts to rehabilitate this testimony by arguing that during cross-examination, Mr. Chess identified "inefficiencies" that

actually “provide a PHOSITA motivation to combine the presented prior art references more efficiently,” but Petitioner fails to point to any testimony or other evidence of record to support this attorney argument.¹⁷ Reply 14.

Petitioner also argues that “Mr. Chess indicated the prior art references surrounded systems with similar components, processes, and operative purposes, and could benefit from combination to create a more desirable product through more thoughtful assembly and piping to create more efficiency.” Reply 18 (citing Ex. 2009 (Chess Depo. Tr.), 113:6–115:8). However, even if Petitioner accurately characterizes the cited testimony, it still “does not supply the details missing from [Mr. Chess’s] Declaration and the Petition as to why a POSA would be motivated to combine Britt or Buese with Hebert.” Sur-reply 13.

On Reply, Petitioner quotes case law indicating that “the teaching, suggestion, or motivation test is flexible,” and a “motivation to combine may be implicit.” Reply 12–13 (citing MPEP 2143 G; *DyStar Textilfarben GmbH & Co. Deutschland KG v. C.H. Patrick Co.*, 464 F.3d 1356, 1366 (Fed. Cir. 2006)). Petitioner also states that “the Federal Circuit has held that ‘a challenger need not prove that there was a known problem with the prior art in order to demonstrate that there was a motivation to combine prior art references.’” *Id.* at 16 (quoting *Arctic Cat Inc. v. Polaris Indus., Inc.*,

¹⁷ Petitioner also argues that Patent Owner’s counsel deliberately attempted to confuse Mr. Chess by questioning him only about figures from the prior art references instead of asking about the references as a whole. See Reply 13–14. Petitioner, however, was free to question Mr. Chess on redirect to refocus his testimony, but on this record does not point us to any redirect or clarifying testimony in this regard. See, e.g., 37 C.F.R. § 42.53(c)(1) (permitting two hours for redirect examination).

795 F. App'x 827, 833 (Fed. Cir. 2019)). While these are all accurate statements of the law, they do little to articulate any specific reason to combine Hebert with Britt or Buese—which the law also requires. *See, e.g., KSR Int'l Co.*, 550 U.S. at 418 (teaching that obviousness requires “determin[ing] whether there was an apparent reason to combine the known elements in the fashion claimed by the patent at issue”).

Petitioner also argues on Reply that “the combination of the references would result in a more efficient product and process, capable of running continuously, and allows for ongoing maintenance and replacement of components and elements without requiring down-time.” Reply 16–17. Patent Owner correctly notes that this discussion is missing from the Petition as a reason to combine Hebert with Britt or Buese. *See* Sur-reply 11–12. Petitioner also nowhere elaborates on exactly how or why the proposed combinations would have been more efficient, been capable of running continuously, or would have minimized down-time, let alone with citation to evidence of record. *See id.*; *see also Johnston v. IVAC Corp.*, 885 F.2d 1574, 1581 (Fed. Cir. 1989) (stating that attorney argument is “no substitute for evidence”).

As discussed above, the Petition asserts that a person of ordinary skill in the art would have looked to Hebert to inform possible improvements or alternative approaches to Britt and Buese. *See, e.g.,* Pet. 15. When asked at oral argument what “improvement” would flow from following Hebert’s teaching to soak the source material in the solvent for a predetermined amount of time, Petitioner’s counsel stated that the improvement would be applying a more analytical mindset to the extraction process. Tr. 45:1–18. That rationale, however, is not stated in the Petition.

Petitioner's counsel also appeared to suggest that the concept of soaking the source material in the solvent for a predetermined amount of time is inherent in the flow rate of a process. *See id.* Although that rationale has some appeal, on this record we cannot credit it, because as counsel candidly agreed, it is not articulated in the Petition. *See id.*; *see also id.* at 36:1–25 (Patent Owner's counsel correctly arguing that Petitioner (i) used Hebert as a motivating reference (as opposed to an evidentiary reference), and (ii) did not make an inherency argument).

We note that any necessary modification of Britt or Buese with Hebert (to the extent Hebert is even needed at all) to reach the “predetermined period of time” limitation, particularly under our construction of this term as meaning a “period of time determined in advance,” may have been trivial or a product of common sense. Petitioner, however, is “master of its complaint.” *SAS Inst., Inc. v. Iancu*, 138 S. Ct. 1348, 1355 (2018). Petitioner elected to contend that neither Britt nor Buese teaches this limitation, and to argue that “Hebert provides a rationale for a PHOSITA to expose the source material to the solvent for a predetermined period of time.” *See* Pet. 14. It is thus incumbent on Petitioner to prove up its arguments by a preponderance of the evidence, and as our reviewing court has explained, “[i]t would . . . not be proper for the Board to deviate from the grounds in the petition and raise its own obviousness theory.” *Sirona Dental Sys. GmbH v. Institut Straumann AG*, 892 F.3d 1349, 1356 (Fed. Cir. 2018); *see also In re Magnum Oil Tools Int’l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (“[T]he Board must base its decision on arguments that were advanced by a party, and to which the opposing party was given a chance to respond.”).

On the full trial record, Petitioner fails to adequately explain its proposed combinations of Hebert with Britt or Buese. The Petition does little more than identify Hebert as purportedly teaching the “predetermined period of time” limitation that it contends is missing from Britt and Buese. However, the “[d]etermination of obviousness cannot be based on the hindsight combination of components selectively culled from the prior art to fit the parameters of the patented invention.” *ATD Corp. v. Lydall, Inc.*, 159 F.3d 534, 546 (Fed. Cir. 1998). An obviousness determination also requires a reason to combine, which is wholly missing from the record. *See KSR Int’l Co.*, 550 U.S. at 418 (stating that obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness”). This failure to articulate a reason to combine infects all six of Petitioner’s grounds. *See* PO Resp. 24 (“Petitioner fails to support a motivation to combine either Britt or Buese with Hebert in all six grounds.”). Accordingly, for the reasons discussed above, we determine that Petitioner has not demonstrated by a preponderance of the evidence that the challenged claims are unpatentable as obvious.

III. CONCLUSION

Based on the information presented, we conclude that Petitioner has not demonstrated by a preponderance of the evidence that claims 1–19 are unpatentable.

In summary:

Claims	35 U.S.C. §	Reference(s) /Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
14, 15	103	Britt, Hebert		14, 15
19	103	Britt, Hebert, Buese		19
16	103	Britt, Hebert, Liebert		16
1–8, 17, 18	103	Buese, Hebert, Britt, Main		1–8, 17, 18
9–12	103	Buese, Hebert, Main		9–12
13	103	Buese, Hebert, Main, Jones		13
Overall Outcome				1–19

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has not demonstrated by a preponderance of the evidence that claims 1–19 of U.S. Patent 9,145,532 B2 are unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of this Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2022-00832
Patent 9,145,532 B2

For PETITIONER:

Clement Hayes
Mark Emde
BLOCK 45 LEGAL
clement@block45legal.com
mark.emde@block45legal.com

For PATENT OWNER:

Ryan R. Owens
SPEARHEAD LEGAL LLP
ryan.owens@spearheadlegal.com

Daniel C. Pierron
WIDERMAN MALEK, PL
dpierron@uslegalteam.com